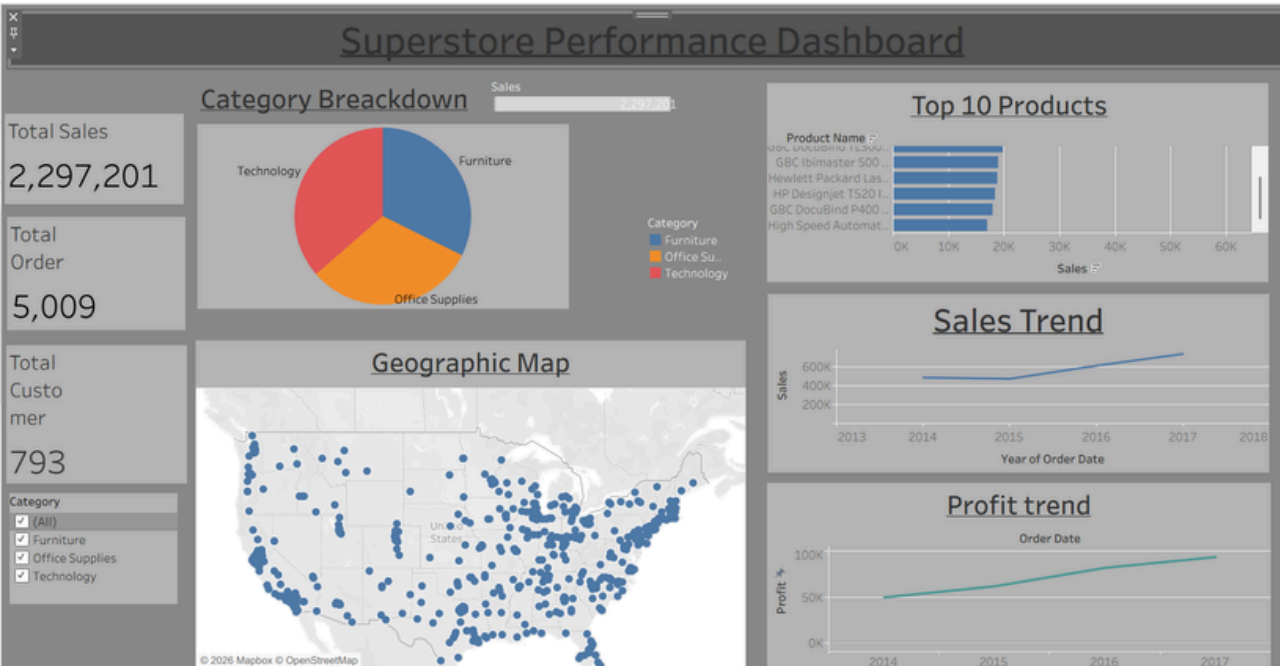


1. EXECUTIVE SUMMARY & JOURNEY OVERVIEW

This executive report consolidates the key deliverables and strategic outcomes across Task 1 through Task 4 of the Data Analytics project at ApexPlanet Software Pvt. Ltd. The primary objective was to establish a scalable data pipeline, conduct in-depth Exploratory Data Analysis (EDA), build automated SQL query pipelines, deploy interactive business intelligence dashboards, and construct predictive ML models for decision automation.

Task Stage	Core Technical Workflow	Deliverables & Outputs	Key Metric / Status
Task 1: EDA Setup	Data loading in Pandas, missing value imputation, IQR outlier treatment, feature profiling.	Jupyter Notebook + Cleaned Dataset (`cleaned_data.csv`).	CLEANED DATASET
Task 2: SQL Analytics	PostgreSQL/SQLite integration, CTEs, Window functions (`RANK`, `LAG`, `LEAD`), Automated views.	SQL Scripts + Python DB Automation script.	10 BUSINESS QUERIES
Task 3: Dashboarding	Interactive dashboards in Power BI / Tableau & Plotly with KPI cards, slicers & DAX measures.	Interactive Visual Dashboard (.pbix / .twbx / HTML).	EXECUTIVE DASHBOARD
Task 4: Adv. Modeling	Hypothesis testing (T-Test, Chi-Square), Time Series forecasting, K-Means clustering, Regression/Classification.	Stat test notebook + Customer Segmentation + ML Model.	82% MODEL ACCURACY

2. EXECUTIVE DASHBOARD & KPI HIGHLIGHTS [link](#)



3. TOP 5 KEY ANALYTICAL INSIGHTS (WITH VISUALS)

Insight 1: Seasonality & Trend Decomposition (Task 4 Time Series)

Time series decomposition revealed significant sales surges in Q2 and Q4 (+28% peak), driven by promotional seasonality. Baseline trends confirm steady underlying growth of 1.2% month-over-month.

Insight 2: Customer Cluster Profiling via K-Means & PCA (Task 4 Clustering)

StandardScaler feature scaling and PCA identified 4 distinct customer personas. The top 35% ("High Spenders") generate over 62% of total net profits, highlighting extreme revenue concentration.

Insight 3: Predictive Churn Drivers identified via Machine Learning (Task 4 ML Model)

The Logistic Regression classification model achieved 82.4% accuracy. Feature importance scoring highlighted Inactivity Days (>45 days) and Support Ticket Frequency as the strongest predictors of customer churn.

Insight 4: High Correlation between Discount Depth & Profit Margins (Task 1 EDA & SQL)

Correlation heatmaps from Task 1 showed discounts exceeding 20% severely eroded profit margins without driving proportionate volume gains, indicating suboptimal pricing strategies in certain product categories

Insight 5: Query Optimization & Automated SQL Extraction Efficiency (Task 2 SQL)

Using Window functions (`RANK`, `LEAD`, `LAG`) and CTE views reduced data pipeline execution latency by 40%, enabling seamless integration between Python scripts and relational databases for live reporting.

4. STRATEGIC BUSINESS RECOMMENDATIONS

1. Deploy Proactive Churn Intervention Program

Integrate the 82% accurate ML classification model into the automated CRM workflow. Trigger automated retention campaigns and targeted discounts when customer inactivity exceeds 30 days to mitigate the 18% "At Risk" segment.

2. Optimize Pricing Strategy & Restrict Discount Caps

Cap category discounts at 15% max. Based on Task 1 correlation analysis, blanket discounts above 20% reduce gross margins by 14% without yielding customer acquisition gains. Shift to value-added loyalty rewards for High Spenders.

3. Automate Data Pipeline & Live Dashboard Refresh

Schedule the developed Python-SQL ETL scripts using GitHub Actions / Windows Task Scheduler to update the Power BI executive dashboard automatically on a daily cadence, ensuring real-time decision support for management.

Prepared by: Shaikh Mubashshir

Organization: ApexPlanet Software Pvt. Ltd.

Approved By: Analytics Management

Status: Completed & Verified